AP BIOLOGY TOPIC 4.5

The Cell Cycle: Interphase (G_1, S, G_2, G_0), Mitosis (PMAT) & Cytokinesis

UNIT 4: CELL COMMUNICATION & CELL CYCLE (10–15% EXAM WEIGHT)

IST-1.B | IST-1.C | GENOME TRANSMISSION

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§1. KEY TERMS & FORMAL DEFINITIONS LEXICON — TOPIC 4.5 (THE CELL CYCLE: INTERPHASE (G_1, S, G_2, G_0), MITOSIS (PMAT) & CYTOKINESIS)

CED Core Vocabulary

Purpose of the Eukaryotic Cell Cycle — Highly regulated sequence of cell growth, DNA replication, and division that produces two genetically identical diploid (2n) daughter cells for organismal...

Mitotic spindle — (microtubules) forms as centrosomes migrate to opposite poles; spindle fibers attach to kinetochore protein plaques on centromeres.

3. Anaphase (Shortest Mitotic Phase) — Separase enzyme cleaves cohesin proteins holding sister chromatids together → sister chromatids separate .

Two nuclear envelopes reform — around the daughter nuclei; nucleoli reappear; mitotic spindle depolymerizes.

1. Prophase (& Prometaphase) — Loosely coiled chromatin condenses into discrete, visible chromosomes (each made of 2 sister chromatids joined at the centromere).

2. Metaphase — Kinetochore microtubules tug chromosomes into a single-file line along the cell equator (the metaphase plate).

4. Telophase — Two complete sets of chromosomes reach opposite poles and decondense back into diffuse chromatin .

Animal Cells (Cleavage Furrow) — A contractile ring of actin microfilaments and myosin motor proteins pinches the plasma membrane inward from the outside until the cell is cleaved in two.

§2. CELL CYCLE OVERVIEW & INTERPHASE (G_1, S, G_2)

~90% of Cell Cycle

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

Cell Cycle Overview & Interphase (G_1, S, G_2) ➔

Purpose of the Eukaryotic Cell Cycle ➔ **G_0 Phase (Non-Dividing / Quiescent State)**

- **Purpose of the Eukaryotic Cell Cycle:** Highly regulated sequence of cell growth, DNA replication, and division that produces **two genetically identical diploid (2n) daughter cells** for **organismal growth, tissue repair/regeneration, and asexual reproduction**.
- **Interphase (~90% of Cell Duration — Longest & Most Metabolically Active Stage):**
  - **G_1 Phase (First Gap):** Cell is highly metabolically active; grows in volume, synthesizes proteins/RNAs, and **duplicates cytoplasmic organelles** (mitochondria, ER, ribosomes).
  - **S Phase (Synthesis): DNA replication occurs!** DNA remains loosely packed as **chromatin**. Each replicated chromosome now consists of **two identical sister chromatids joined at the centromere** by cohesin proteins.
  - **G_2 Phase (Second Gap):** Final preparation for mitosis — rapid synthesis of microtubules/spindle proteins, high **ATP production**, and **centrosome duplication** completes.
- **G_0 Phase (Non-Dividing / Quiescent State):**
  - Cells exit the active cell cycle at **G_1** and do not divide, while remaining **fully alive and metabolically active** performing specialized physiological functions.
  - **Permanent G_0:** Mature human **neurons** and **skeletal/cardiac muscle cells** never divide again.
  - **Temporary G_0: Liver cells (hepatocytes)** and lymphocytes rest in **G_0** but can **re-enter G_1** when stimulated by external growth factors (e.g., after liver injury).

[COMMON AP EXAM FRQ TRAP] **3 High-Frequency Exam Traps on Interphase & G_0:**

1. **Interphase is NOT a "resting phase"** — the cell is furiously synthesizing DNA, RNA, proteins, and organelles!
2. **G_0 does NOT mean dead or metabolically inactive** — neurons in **G_0** consume massive ATP firing action potentials!
3. **Chromosome number (2n = 46) does NOT double in S phase!** Only **DNA mass/content (2C → 4C) and chromatid count (46 → 92) double** because sister chromatids sharing 1 centromere still count as **1 chromosome**!

§4. MITOSIS (M PHASE: PMAT) & CYTOKINESIS

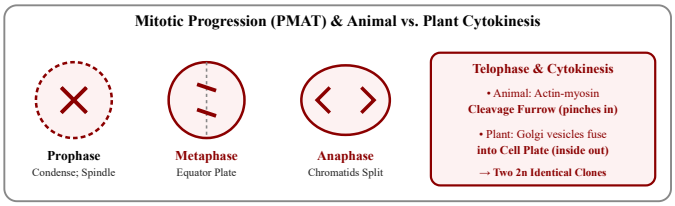
Equal Genome Division

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

Mitosis (M Phase ➔ **1. Prophase (& Prometaphase)** ➔ **2. Metaphase** ➔ **3. Anaphase (Shortest Mitotic Phase)** ➔ **4. Telophase**

- **1. Prophase (& Prometaphase):**
  - Loosely coiled chromatin **condenses into discrete, visible chromosomes** (each made of 2 sister chromatids joined at the centromere).
  - Nucleolus disappears and **nuclear envelope fragments**.
  - **Mitotic spindle** (microtubules) forms as centrosomes migrate to opposite poles; spindle fibers attach to **kinetochore** protein plaques on centromeres.
- **2. Metaphase:**
  - Kinetochore microtubules tug chromosomes into a single-file line along the cell equator (the **metaphase plate**).
  - *M-checkpoint verifies every kinetochore is attached to opposite poles before proceeding.*
- **3. Anaphase (Shortest Mitotic Phase):**
  - Separase enzyme cleaves cohesin proteins holding sister chromatids together → **sister chromatids separate**.
  - Kinetochore microtubules depolymerize (shorten) to **pull daughter chromosomes to opposite poles**, while non-kinetochore microtubules elongate the cell.
- **4. Telophase:**
  - Two complete sets of chromosomes reach opposite poles and **decondense back into diffuse chromatin**.
  - **Two nuclear envelopes reform** around the daughter nuclei; nucleoli reappear; mitotic spindle depolymerizes.
- **5. Cytokinesis (Division of the Cytoplasm — Overlaps Late Telophase):**
  - **Animal Cells (Cleavage Furrow):** A contractile ring of **actin microfilaments and myosin motor proteins** pinches the plasma membrane inward from the outside until the cell is cleaved in two.
  - **Plant Cells (Cell Plate):** Because plants have a rigid cell wall, **Golgi-derived vesicles** carrying cell-wall polysaccharides coalesce at the midline to build a **cell plate** from the inside out, fusing with the plasma membrane to form a new cell wall.

[COMMON AP EXAM FRQ TRAP] **Terminology Precision in Anaphase:** The exact moment sister chromatids separate at the centromere in **Anaphase**, each chromatid is immediately considered a full, independent **individual chromosome**!



§3. TRACKING CHROMOSOMES (N) VS. DNA MASS (C) IN HUMANS

Math / Graph Reference

CELL CYCLE STAGE	PLOIDY / CHROMOSOMES	SISTER CHROMATIDS	RELATIVE DNA CONTENT (C)
G_1 (or G_0)	**2n = 46** (unreplicated)	0 (46 single molecules)	**2C** (baseline diploid mass)
After S / G_2 / Pro / Meta	**Still 2n = 46** (replicated)	**92 sister chromatids**	**4C (Doubled in S phase!)**
Anaphase / Telophase *(in 1 cell before cleavage)*	Temporarily **4n = 92** (once centromeres split!)	0 (now individual chromosomes)	**4C** (**2C** at each pole)
After Cytokinesis *(per daughter cell)*	**2n = 46** (identical to parent)	0	**2C** (halved back to baseline)

§5. SCHOLARLY SYNTHESIS: EXPERIMENTAL DESIGN, QUANTITATIVE ANALYSIS, PATHOLOGY & AP FRQ RHETORIC

High-Yield Exam Rigor

MASTER TOPIC SYNTHESIS & CAUSAL ARCHITECTURE — TOPIC 4.5 **[MOLECULAR CAUSE ➔ CELLULAR MECHANISM ➔ PHYSIOLOGICAL / ECOLOGICAL EFFECT ➔ FRQ JUSTIFICATION]**

Topic 4.5: The Cell Cycle: Interphase (G_1, S, G_2, G_0), Mitosis (PMAT) & Cytokinesis ➔ **Structural / Energetic Determinants** ➔ **Regulated Pathway & Homeostasis** ➔ **Experimental Perturbation (±2 SE_x̄)** ➔ **Phenotypic / Evolutionary Outcome**

SCHOLARLY DIMENSION	COMPREHENSIVE ACADEMIC REFERENCE & MECHANISTIC SPECIFICATION
I. Experimental Design & Required Lab Controls	AP Investigation 7 (Mitosis in Onion Root Tips & Lectin Signaling): Root tip squash stained with aceto-orcein/toluidine blue to compare mitotic index (% dividing cells = [cells in PMAT / total cells counted] × 100) between control meristem and phytohemagglutinin (lectin) or caffeine-treated roots. Chi-square (χ^2) tests whether observed mitotic vs. interphase counts differ significantly from control. **Null Hypothesis (H_0):** State explicitly that the independent variable treatment has *no statistically significant effect* on the dependent variable; any observed difference is due to random sampling chance alone.
II. Quantitative Formulas & Graph Interpretation	Quantitative & Graph Analysis: Mitotic Index = (N_mitosis / N_total) × 100. DNA Content (C) & Ploidy (n) tracking in diploid (2n) somatic cells: G_1 = 2n, 2C (unreplicated chromosomes); S phase = DNA synthesis (2n, 2C → 4C); G_2/Prophase/Metaphase = 2n, 4C (sister chromatids joined); Anaphase = transient 4n, 4C (chromatids separate into individual chromosomes); Post-Cytokinesis = 2n, 2C. **Error Bar Rule (±2 SE_x̄):** If the ±2 SE_x̄ error bars of two treatment groups *do not overlap*, there is a statistically significant difference between means ($p < 0.05$; reject H_0).
III. Pathology, Mutation & Disruption Case Study	Clinical & Signaling Disruption Case Study: Vibrio cholerae secretes Cholera Toxin, which enzymatically modifies the G_α subunit of intestinal GPCRs so it cannot hydrolyze GTP → GDP. Locked in its active GTP-bound state, G_α continuously activates adenylyl cyclase, causing runaway cAMP accumulation, constitutive PKA activation, CFTR Cl^-/H_2O efflux, and life-threatening secretory diarrhea. In cell cycle control, loss-of-function mutations in both alleles of TP53 ('guardian of the genome') prevent p21-mediated G_1/S arrest and apoptosis of DNA-damaged cells, driving oncogenic proliferation.
IV. AP FRQ Task Verb Scoring Templates	• **DESCRIBE / IDENTIFY:** State the direct trend, value, or structural component with exact units from the figure. • **EXPLAIN (How/Why):** Connect cause to effect via a 3-link chain: *"[Change in X] causes [molecular/cellular mechanism Y], which results in [observable outcome Z]."* • **PREDICT:** State the specific directional change (increase, decrease, or remain constant) when the pathway is perturbed. • **JUSTIFY:** Pair quantitative data/evidence from the prompt with the underlying biological principle from Topic 4.5 to prove why the claim holds.

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